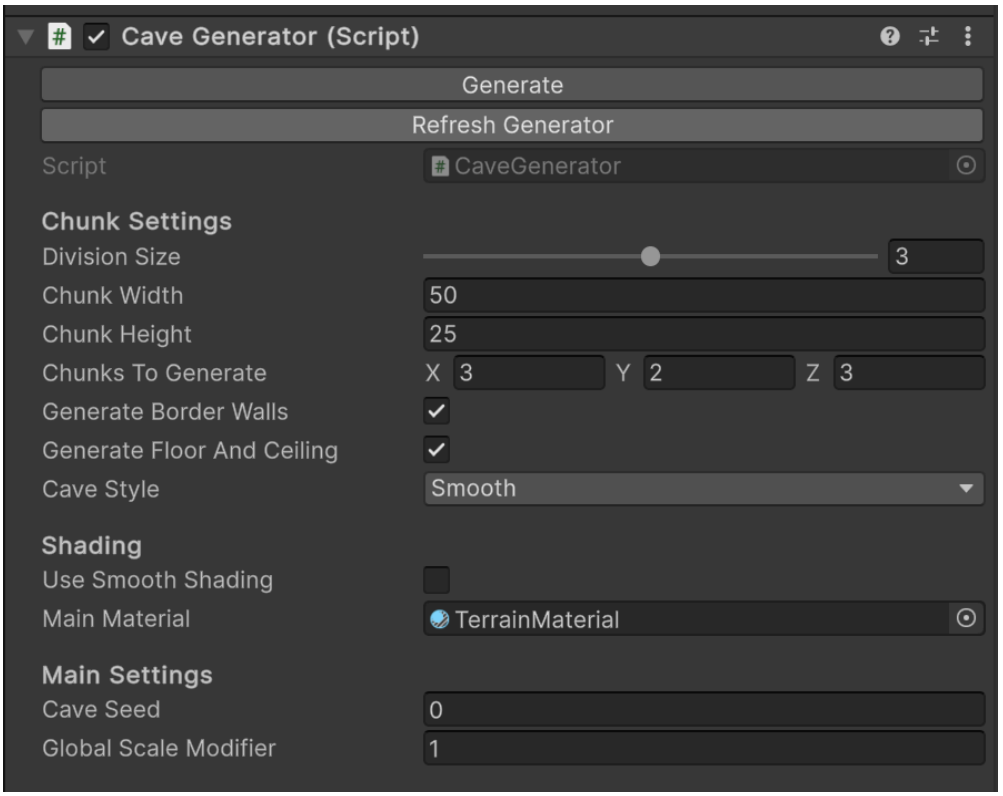
# North Sea Games - Low Poly Cave Generator Asset Summary

A simple setup of the cave generator asset is provided in the demo scene. Below is a summary of the asset, to aid users in creating and modifying caves.

## Basic Settings




Use the "generate" button to generate new caves. **It is best practice to not scale or rotate the cave GameObject** (although it can be translated if you need to).

If the generation is interrupted (eg. by a recompilation), you can use the "refresh generator" button to clear the internal flags (this is needed because the script uses coroutines during edit mode).

**Division size -** resolution of the cave mesh (a smaller value gives more detail)

**Chunk width / height -** size of each individual cave chunk

**Chunks to generate -** number of chunks to generate for each axis

**Generate border walls -** toggles generation of walls to enclose the caves

**Generate floor and ceiling -** toggles generation of the cave ceiling and floor

**Cave style -** "smooth" will apply interpolation to the generation, reducing the blocky look of the caves. "angular" has no interpolation, giving a more pronounced low-poly look. "cubes" generates voxel caves

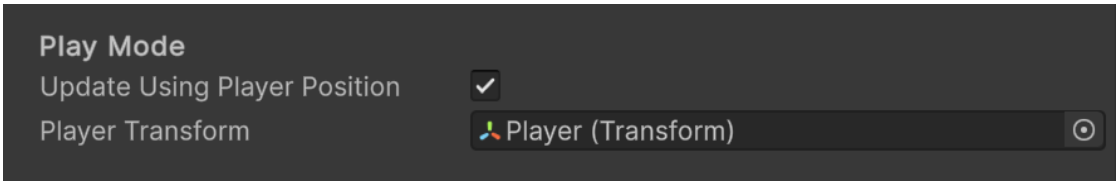
**Use smooth shading -** If set, the cave mesh will be smooth shaded (this takes longer to generate)

**Main Material -** this material will be applied to the cave mesh

**Cave seed -** random seed for generating the caves

**Global scale modifier -** overall scale of the generated caves

## Play Mode

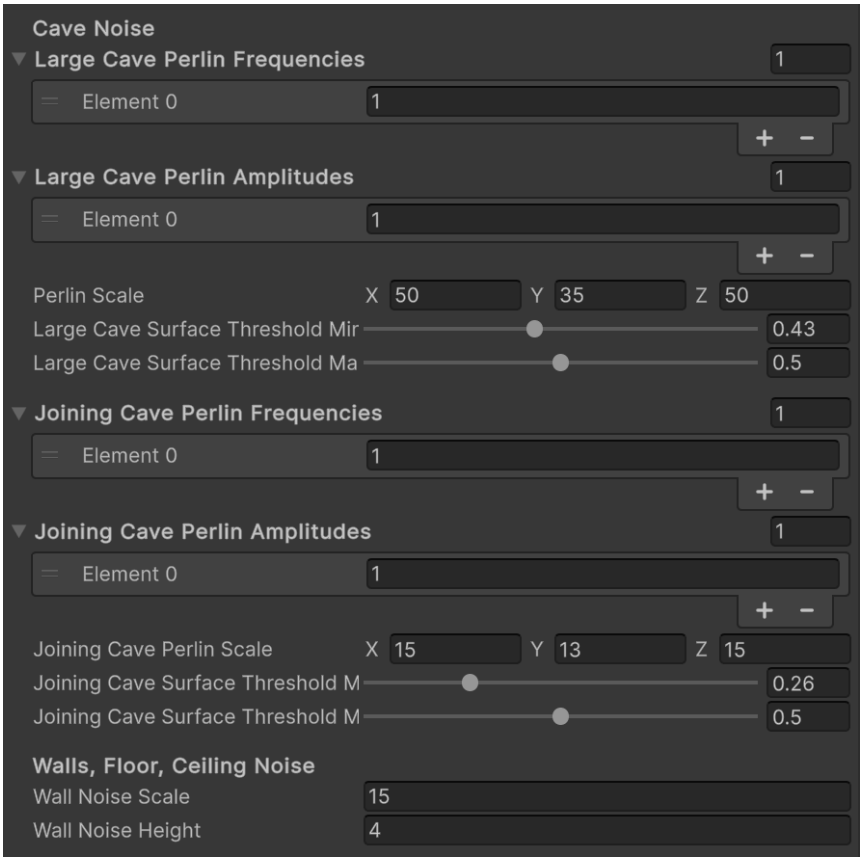



**Update using player position -** enable this to have the caves update dynamically as the player moves around. Cave walls will not be generated if this is enabled

**Player transform -** position of the player for cave update purposes

If you are using this feature, **please make sure the player starts within the bounds of the cave** (otherwise the caves may not update correctly).

## Noise Settings




Caves are formed from two overlapping sets of noise, one "large" which generates large caves, and one "joining" which creates smaller tube-like caves that join the large ones together. You can set any values to use the two sets however you like

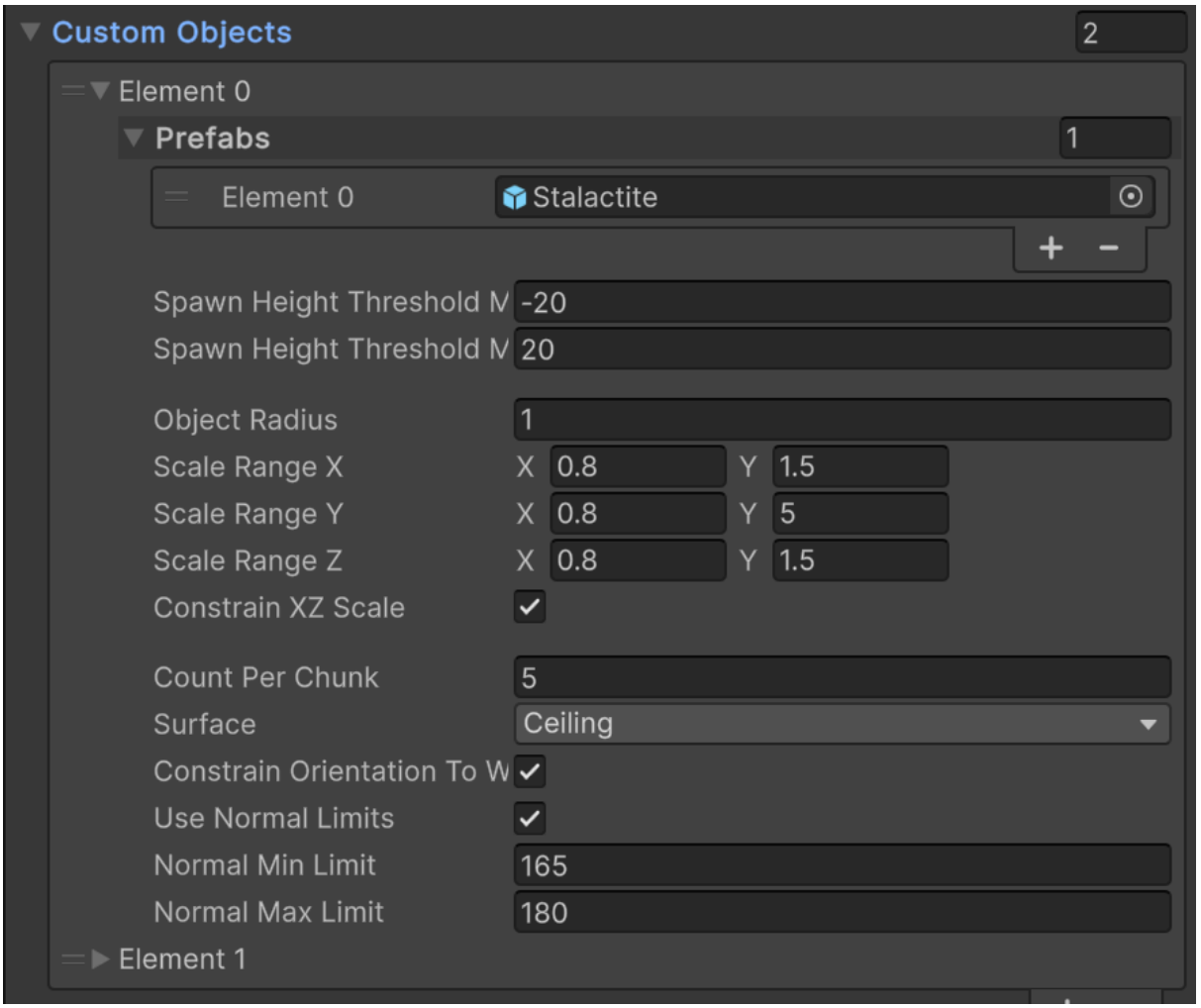
**Perlin frequencies / amplitudes -** these are the base frequencies and amplitudes of the noise that is used for generation. These will affect the overall shape and undulation of the caves

**Perlin scale -** scale of the perlin noise used for generation (separated along each axis). Change this to modify the smoothness of caves

**Min / max surface thresholds -** noise values that are outside of these thresholds define the cave surfaces (i.e what is considered "rock"). These can be adjusted to modify the cave structure.

**Walls, floor, ceiling noise -** these parameters control the size and height of the noise used to generate the walls, ceiling, and floor

## Custom objects




Above shows an example setup for a custom object. The settings are similar for rocks and trees.

**Prefabs -** list of prefabs to select from for this object placement

**Spawn height threshold min -** minimum placement height within the cave

**Spawn height threshold max -** maximum placement height within the cave

**Object radius -** radius of free space required by the objects for valid placement

**Scale ranges -** adjust these to change the size distribution of the objects

**Constrain XZ scale -** if set, the generator will only use the X scale range (use this to keep the objects square)

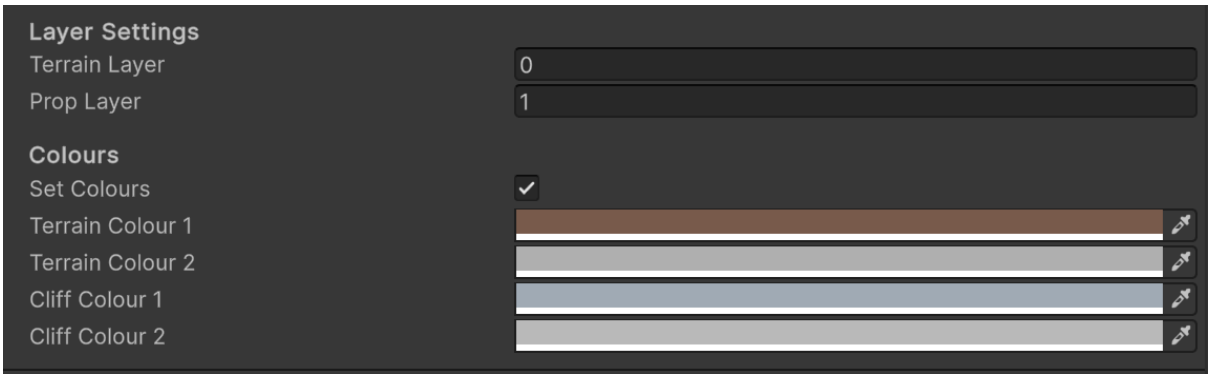
**Count per chunk -** total number of objects to place on each chunk

**Surface -** choose between placing the objects on the ceiling, walls, or floor

**Constrain orientation to world axes -** this constrains objects to face cardinal directions only

**Use normal limits -** if set, objects will only be placed at points which are within the normal limits set below. The normal limits are the angle from the world up vector (y == 1)

## Layers, Colours




**Layers -** Use these to specify the layers for the cave mesh (terrain) and placed objects (props).

**When creating custom prefabs to be placed within the cave, make sure they are in the same layer that is set for props here, and that they have a collider.**

Colours for the caves can also be set here - note that if "set colours" is enabled, it will modify the colour properties of the base material. Disable this if you would prefer to set colours in the material itself (or if you have assigned a custom material).

## Generating at runtime

To generate a new cave during play mode, create a new game object and add the generator script. You can then set appropriate values, and call the Generate() function.

## Prefabs

Cave objects can be saved as prefabs, although the generator will not function correctly when the editor is in prefab mode.

Thanks for using a North Sea Games asset!

**Help and support / contact -** emgamesdev@gmail.com